

Problem Set 2

Q3) The coefficient for volume expansion for an ideal gas is β :

$$\beta = \frac{1}{T}$$

The expression for Grashof number is:

$$Gr = \frac{g\beta(T_s - T_\infty)L^3}{\nu^2}$$

Q8)

Q10)

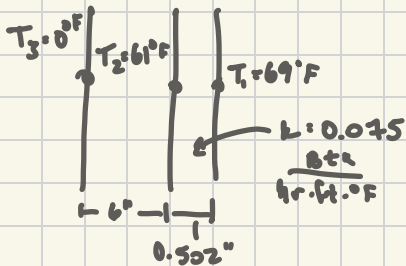
P10A.5)

average velocity expression:

$$\langle v_z \rangle = \frac{(\rho g \beta \Delta T) B^2}{48 \mu}$$

* skip *

P 10.A6)



$$a) \int \frac{\partial}{\partial t} \rho u dV = - \int_S \langle q/n \rangle dS + \int \dot{S} dV$$

$$0 = \int_S \langle q/n \rangle dS$$

$$\frac{dq_x}{dx} = 0 \Rightarrow \frac{d}{dx} \left(-k \frac{dT}{dx} \right) = 0$$

$$\frac{d^2 T}{dx^2} = 0 \quad (\text{linear})$$

$$\frac{dT}{dx} = \frac{T_2 - T_1}{x_2 - x_1}$$

$$q_x = k \frac{T_1 - T_2}{x_2 - x_1}$$

$$q_x = 0.075 \left[\frac{\text{Btu}}{\text{hr} \cdot \text{ft} \cdot ^\circ\text{F}} \right] \left(\frac{69 - 61}{0.502/12} \right) \left[\frac{^\circ\text{F}}{\text{ft}} \right]$$

$$\boxed{q_x = 14.34 \frac{\text{Btu}}{\text{hr} \cdot \text{ft}^2}} \quad \text{yay!}$$

b)

$$q_x \text{ of panel} = q_x \text{ of wall}$$

$$q_{x\text{wall}} = k_{\text{wall}} \frac{T_2 - T_3}{x_3 - x_2}$$

$$k_{\text{wall}} = q_{x\text{wall}} \frac{x_3 - x_2}{T_2 - T_3}$$

$$k_{\text{wall}} = 14.34 \left[\frac{\text{Btu}}{\text{hr} \cdot \text{ft}^2} \right] \left(\frac{6/12}{61 - 0} \right) \left[\frac{\text{ft}}{^\circ\text{F}} \right]$$

$$k_{\text{wall}} = 0.1176$$

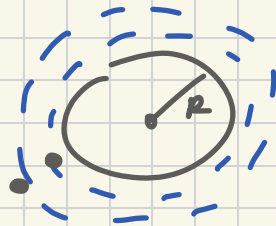
$$\text{"thermal resist."} = \frac{x_3 - x_2}{k_{\text{wall}}}$$

$$= \frac{0.5}{0.1176} \text{ ft} \cdot \frac{\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}}{\text{Btu}} = \boxed{4.25 \frac{\text{ft}^2 \cdot \text{hr} \cdot ^\circ\text{F}}{\text{Btu}}}$$

P 10B.1)

a) $T = T(r)$

$$q_r = -k \frac{dT}{dr}$$



$$0 = - \int_S \langle q/n \rangle dS$$

surf. A. of sphere:
 $4\pi r^2$

$$\frac{[q_r \cdot \cancel{4\pi} r^2]_r - [q_r \cdot \cancel{4\pi} r^2]_{r+\Delta r}}{4\pi \Delta r} = 0$$

$$- \frac{d}{dr} (r^2 q_r) = 0$$

$$= - \frac{d}{dr} \left(r^2 \cdot -k \frac{dT}{dr} \right) = 0$$

$$= \boxed{k \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0}$$

b) $\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$

$$r^2 \frac{dT}{dr} = C_1$$

$$\frac{dT}{dr} = \frac{C_1}{r^2} \quad \int dT = \int C_1 r^{-2} dr$$

$$T = C_2 - \frac{C_1}{r}$$

B.C.:

① @ $r=R \rightarrow T=T_R$

② @ $r=\infty \rightarrow T=T_\infty$

$$T_\infty = C_2 - \frac{C_1}{(r=\infty)} \Rightarrow C_2 = T_\infty$$

$$T_R = T_\infty - \frac{C_1}{R}$$

$$T_\infty - T_R = \frac{C_1}{R} \Rightarrow C_1 = R(T_\infty - T_R)$$

so, $\boxed{T(r) = T_\infty - \frac{R}{r}(T_\infty - T_R)}$

c) $q_r = -k \frac{dT}{dr}$

$$\frac{dT}{dr} = \frac{R}{r^2} (T_\infty - T_R)$$

$$q_r = -\frac{R}{r^2} (T_\infty - T_R) k$$

$$q_r(r=R) = -\frac{R}{R^2} (T_\infty - T_R) k$$

$$\boxed{q_{r, \text{surface}} = \frac{k}{R} (T_R - T_\infty)}$$

Newton's Law of Cooling:

$$Q_x = h_a(T_a - T_o)$$

so,

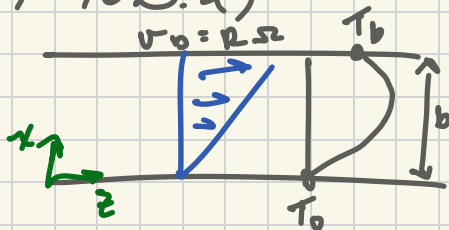
$$h = \frac{k}{R}$$

$$Nu = \frac{hD}{k} = \frac{\cancel{k}}{\cancel{k}} \cdot 2\cancel{k} \cdot \frac{1}{\cancel{k}} = 2$$

d) Nusselt number (Nu): compares convection @ surface to conduction in fluid

Biot number (Bi): compares external convection resistance to internal inside a solid.

P 10B.2)



internal energy balance:

$$0 = \int_S \langle q/n \rangle dS + \int_V |\rho| |\gamma_s| dV$$

$$0 = \cancel{[LWq_x]}_x - \cancel{[LWq_x]}_{x+\Delta x} + \mu \left(\frac{dv_z}{dx} \right)^2 (\cancel{LW\Delta x})$$

$\cancel{LW\Delta x}$

$$0 = - \frac{dq_x}{dx} + \mu \left(\frac{dv_z}{dx} \right)^2$$

$$\star v_z = \frac{v_b}{b} x \Rightarrow \frac{dv_z}{dx} = \frac{v_b}{b}$$

$$\frac{dq_x}{dx} = \mu \left(\frac{v_b}{b} \right)^2$$

B.C.: (1) $x=0, T=T_0$ (2) $x=b, q_x=0$

$$q_x = \mu \left(\frac{v_b}{b} \right)^2 x + C_1$$

$$(2) \quad 0 = \mu \left(\frac{v_b}{b} \right)^2 b + C_1$$

$$\Rightarrow C_1 = - \mu \left(\frac{v_b}{b} \right)^2 b$$

$$q_x = \mu \left(\frac{v_b}{b} \right)^2 x - \mu \left(\frac{v_b}{b} \right)^2 b$$

$$\frac{dT}{dx} = -\frac{\mu}{k} \left(\frac{v_b}{b} \right)^2 x + \frac{\mu}{k} \left(\frac{v_b}{b} \right)^2 b$$

$$T = -\frac{\mu}{k} \left(\frac{v_b}{b} \right)^2 \cdot \frac{x^2}{2} + \frac{\mu}{k} \left(\frac{v_b}{b} \right)^2 b x + C_2$$

$$\textcircled{1} T_0 = C_2$$

$$T = \frac{\mu}{k} \left(\frac{v_b}{b} \right)^2 \left(b x - \frac{x^2}{2} \right) + T_0$$

$$\boxed{\frac{k(T - T_0)}{\mu v_b^2} = \frac{x}{b} - \frac{1}{2} \left(\frac{x}{b} \right)^2}$$

↳ Book's Answer:

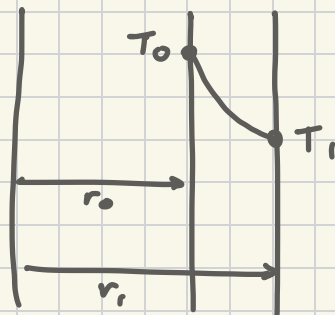
$$\frac{T - T_0}{\mu v_0^2 / k} = \left(\frac{x}{b} \right) - \frac{1}{2} \left(\frac{x}{b} \right)^2$$

where did v_0 come from?

P 10B.4)

$$a) \quad \left. \begin{aligned} k &= k_0 (T = T_0) \\ k &= k_1 (T = T_1) \end{aligned} \right\} \text{linear}$$

$$0 = \int_s \langle q | n \rangle$$



$$\frac{[2\pi r \cancel{q_r}]_r - [2\pi r \cancel{q_r}]_{r+\Delta r}}{2\pi \cancel{r} \Delta r} = 0$$

$$\frac{1}{r} \frac{d}{dr} (r q_r) = 0 \Rightarrow \frac{d}{dr} (r q_r) = 0$$

$$r q_r = C_1 \quad -k r \frac{dT}{dr} = C_1$$

$$\frac{dT}{dr} = -\frac{C_1}{k r} \quad T = -\frac{C_1}{k} \ln r + C_2$$

$$\text{B.C.: } T(r_0) = T_0 \quad T(r_1) = T_1$$

$$T_0 = -\frac{C_1}{k} \ln r_0 + C_2$$

$$- \quad T_1 = -\frac{C_1}{k} \ln r_1 + C_2$$

$$T_0 - T_1 = \frac{C_1}{k} \left(\ln \frac{r_1}{r_0} \right)$$

$$C_1 = k (T_0 - T_1) \left(\ln \frac{r_1}{r_0} \right)^{-1}, \quad q_r = \frac{C_1}{r}$$

$$Q_r = \frac{k}{r} (T_0 - T_1) \left(\ln \frac{r_1}{r_0} \right)^{-1}$$

@ $r = r_0$

$$Q_r = \frac{k}{r_0} (T_0 - T_1) \left(\ln \frac{r_1}{r_0} \right)^{-1} \cdot 2\pi r_0 L$$

$$Q_r = 2\pi L k (T_0 - T_1) \left(\ln \frac{r_1}{r_0} \right)^{-1}$$

use "average k ":

$$k_{avg} = \frac{k_0 + k_1}{2}$$

$$\boxed{Q_r = 2\pi L (T_0 - T_1) \left(\frac{k_0 + k_1}{2} \right) \left(\ln \frac{r_1}{r_0} \right)^{-1}}$$

b) when $(r_1 - r_0)/r_0$ is very small:

$$\delta = r_1 - r_0 \ll r_0$$

$$\frac{r_1}{r_0} = \frac{\delta + r_0}{r_0} = \frac{\delta}{r_0} + 1$$

$$\ln \left(\frac{r_1}{r_0} \right) = \ln \left(\frac{\delta}{r_0} + 1 \right) \approx \frac{\delta}{r_0} = \frac{r_1 - r_0}{r_0}$$

$$Q_r = 2\pi L (T_0 - T_1) \left(\frac{k_0 + k_1}{2} \right) \left(\frac{r_0}{r_1 - r_0} \right)$$

$$\Rightarrow \boxed{Q_r = 2\pi L r_0 \left(\frac{k_0 + k_1}{2} \right) \left(\frac{T_0 - T_1}{r_1 - r_0} \right)}$$

P 10B.7)

$$a) \rho C_p v_z(x) \frac{\partial T}{\partial z} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} \right) + \cancel{q}$$

$$\Rightarrow \boxed{\rho C_p v_z(x) \frac{\partial T}{\partial z} = k \frac{\partial^2 T}{\partial x^2}}$$

$$b) \theta = \frac{T - T_1}{T_0 B/k} \quad \eta = \frac{x}{B} \quad \zeta = \frac{k z}{\rho \hat{C}_p v_{z\max} B^2}$$

$$v_z(x) = v_{\max} \left(1 - \frac{\eta^2}{B^2} \right)$$

$$\frac{\partial}{\partial z} = \frac{k}{\rho \hat{C}_p v_{\max} B^2} \frac{\partial}{\partial \zeta}$$

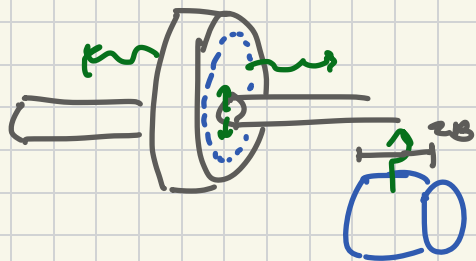
$$\frac{\partial^2}{\partial x^2} = \frac{1}{B^2} \frac{\partial^2}{\partial \eta^2}$$

$$\frac{v_z}{v_m} = 1 - \eta^2$$

$$\Rightarrow \boxed{(1 - \eta^2) \frac{\partial \theta}{\partial \eta} = \frac{\partial^2 \theta}{\partial \eta^2}}$$

P 10D.1)

$$a) \quad \dot{Q} = \int_S \{q/n\} dS$$



$$\frac{[4\pi r q_r]_r - [4\pi r q_r]_{r+\Delta r} - 2\pi \Delta r^2 h (T - T_a)}{4\pi r \Delta r} = 0$$

$$\frac{1}{r} \frac{d}{dr} (r q_r) - \frac{h}{2\pi r \Delta r} (T - T_a) = 0$$

$$\frac{1}{r} \frac{d}{dr} \left(-k r \frac{dT}{dr} \right) - \frac{h}{2\pi r \Delta r} (T - T_a) = 0$$

$$\cancel{\frac{d}{dr}} \left(r \frac{dT}{dr} \right) - \frac{h}{2\pi \cancel{\Delta r} k} (T - T_a)$$

$$r \frac{dT}{dr} = \frac{h}{2\pi k} (T - T_a)$$

$$\frac{dT}{dr} \cdot \frac{h}{2\pi k} (T - T_a) \frac{1}{r}$$

$$T = \frac{h}{2\pi k} (T - T_a) \ln r + C_1$$

$$\text{B.C.: } T = T_0 \text{ @ } r = R_0$$

$$T_0 = \frac{h}{2\pi k} (T_0 - T_a) \ln R_0 + C_1$$

$$C_1 = T_0 - \frac{h}{2\pi k} (T_0 - T_a) \ln R_0$$

$$T - T_0 = \frac{h}{2\pi k} \left[(T - T_a) \ln r - (T_0 - T_a) \ln R_0 \right]$$